

Discrete Mathematics

Cardinality of Sets and Infinity

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What Is Cardinality?

Cardinality measures the “size” of a set.

For finite sets, this is simply the number of elements:

$$|\{1, 2, 3\}| = 3$$

However, for infinite sets, “size” becomes much less intuitive. This section introduces a rigorous way to compare the sizes of infinite sets using functions (bijections, injections).

Cardinality: Formal Definition

Two sets A and B have the same cardinality if there exists a **one-to-one correspondence** (a bijection) between them:

$$|A| = |B|$$

Bijection =

- One-to-one (injective)
- Onto (surjective)

This means:

- Every element of A matches uniquely with one element of B ,
- No leftover elements on either side.

Comparing Cardinalities

If there exists an injective function $f : A \rightarrow B$, we write:

$$|A| \leq |B|$$

This means:

- No two elements of A map to the same element in B ,
- A fits inside B (possibly with room to spare).

If additionally A and B do **not** have the same cardinality, then:

$$|A| < |B|$$

Countable Sets

A set is **countable** if:

- It is finite, or
- It has the same cardinality as the positive integers $\mathbb{Z}^+$

Countably infinite sets all share the cardinality:

$$|\mathbb{Z}^+| = \aleph_0 \quad (\text{aleph-null})$$

Examples of countable sets:

- Positive integers
- All integers
- Even integers
- Rational numbers (surprisingly!)

Uncountable Sets

A set is **uncountable** if it is not countable.

Most famous example:

$\mathbb{R}$ (the real numbers)

The real numbers cannot be listed in a sequence indexed by the positive integers.

This means:

$$|\mathbb{R}| > |\mathbb{Z}^+|$$

How Do We Show a Set Is Countable?

An infinite set is countable if and only if:

It is possible to list its elements in a sequence (indexed by the positive integers).

This means:

$$a_1, a_2, a_3, \dots$$

Every countably infinite set can be placed into such a list, even if the structure of the set is complex.

Why Does Listing Work?

If $f : \mathbb{Z}^+ \rightarrow S$ is a bijection, then:

$$a_1 = f(1), a_2 = f(2), a_n = f(n)$$

So listing a set's elements is equivalent to constructing a bijection from positive integers.

This creates a **one-to-one correspondence** between the infinite set and $\mathbb{Z}^+$.

Hilbert's Grand Hotel (Part 1)

Hilbert imagined a hotel with **countably infinite rooms**: Room 1, Room 2, Room 3, ...

Each room is occupied, yet a new guest arrives.

Surprisingly, the hotel can still accommodate the guest.

This illustrates the strange behavior of infinite sets.

Hilbert's Grand Hotel (Part 2)

To make room:

- Move guest in Room 1 $\rightarrow$ Room 2
- Room 2 $\rightarrow$ Room 3
- Room 3 $\rightarrow$ Room 4
- ...
- Room $n \rightarrow$ Room $n + 1$

This frees up Room 1 for the new guest.

Key idea: Even though the hotel is “full,” shifting the infinite list creates space.

This is impossible for finite sets but works for infinite ones.

Example 1: Positive Even Integers Are Countable

We want to show the set:

$$E = \{2, 4, 6, 8, \dots\}$$

is countable.

Define:

$$f(n) = 2n$$

Why is this a bijection?

One-to-one: If $2n = 2m$, then $n = m$. **Onto:** For every even $t = 2k$, we have $f(k) = t$.

Thus E is countably infinite.

Example 2: The Integers Are Countable

We want to list:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

A valid sequence is:

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Or define:

$$f(n) = \begin{cases} n/2, & n \text{ even} \\ -(n-1)/2, & n \text{ odd} \end{cases}$$

This creates a bijection from $\mathbb{Z}^+$ to $\mathbb{Z}$.

Thus, $\mathbb{Z}$ is countable.

Rational Numbers Are Countable (Definition)

A rational number is:

$$\frac{p}{q} \quad \text{where } p, q \in \mathbb{Z}, \quad q \neq 0$$

Examples:

$$\frac{3}{4}, \quad \frac{7}{1}, \quad \frac{1}{1000}$$

But:

$$\sqrt{2} \notin \mathbb{Q}$$

We now show the positive rationals are countable.

Example 3: Positive Rationals Are Countable

The idea is to arrange the rationals in a 2D grid:

Rows = denominator q Columns = numerator p

We then traverse the grid diagonally:

$$p + q = 2, 3, 4, 5, \dots$$

Sequence begins:

$$1, \frac{1}{2}, 2, 3, \frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \dots$$

Fractions equivalent to earlier values (such as $\frac{2}{2}$, $\frac{3}{3}$) are skipped.

Example 4: Strings Over a Finite Alphabet

Let A be a finite alphabet.

A finite string over A is a sequence like:

$a, b, aa, ab, ba, bb, \dots$

To show strings are countable, list them:

- 1 All strings of length 0 (just λ)
- 2 All strings of length 1 in alphabetical order
- 3 All strings of length 2 in lexicographic order
- 4 All strings of length 3, and so on

This listing gives a bijection from $\mathbb{Z}^+$ to all finite strings.

Thus, the set of finite strings is countably infinite.

Lexicographic Order

Lexicographic order means ordering strings the same way words are ordered in a dictionary.

How it works:

- Compare the first characters of two strings.
- If they differ, the one with the smaller character comes first.
- If they are the same, compare the second characters.
- Continue until a difference is found.
- If all compared characters match but one string is shorter, the shorter one comes first.

Example (alphabet $A = \{a, b\}$):

$a, b, aa, ab, ba, bb, aaa, aab, aba, \dots$

This ordering is used in showing that the set of all finite strings over a finite alphabet is countably infinite, because we can list all strings systematically by increasing length and lexicographic order.